

Supplementary materials of “Internal Feature Interaction Mining for Configuration Prioritization”

I. EVALUATION METRICS

II. ADDITIONAL EXPERIMENTS

A. Classification Performance

To handle RQ4, we sample configurations using different sampling algorithms and apply HCP to these configurations to evaluate fault-detection performance. **To answer RQ5**, we vary the parameters α and β to examine how different weightings of the interaction score, novelty, and diversity affect the performance of HCP. In addition, we conduct an ablation study by omitting individual components of the edge weight to create corresponding HCP variants and comparing their fault-detection performance, thereby assessing the contribution of each component. Finally, we construct the global edge sets using only control interactions (i.e., logical presence conditions) and only data-dependence interactions, respectively, and compare their performance to analyze their impact on HCP.

B. Robustness Analysis (RQ4)

Fig. 1 shows the effect of different sampling algorithms on HCP; panels (a) and (b) report the results for *Rank* and *APFD*, respectively. For the Rank metric, ICPL-based sampling yields the worst average Rank, followed by Chvatal-based sampling. However, Fig. 1(b) indicates that Chvatal- and ICPL-based sampling achieve higher average APFD than Yasa- and IncLing-based sampling. The primary reason is that Chvatal and ICPL produce substantially more configurations than the other two samplers, which, under the same number of detected faults, leads to higher APFD. By contrast, when comparing the Rank and APFD achieved with IncLing and Yasa, IncLing provides a greater improvement for HCP.

We further analyze distributional differences across the configuration sets generated by the four sampling algorithms. Table I summarizes five statistics: the average number of configurations, the Gini coefficient of feature selections, the mean nearest-neighbor Jaccard similarity (NN-Jaccard), the 10th percentile of NN-Jaccard, and the mean configuration size. Specifically, the Gini coefficient is computed from per-feature selection counts across all configurations. NN-Jaccard measures, for each configuration, the Jaccard similarity to its most similar neighbor and then averages these values over the set. The 10th percentile of NN-Jaccard (p10) emphasizes the lower tail of this distribution and highlights the least-similar neighborhoods. The mean configuration size is the average number of selected features per configuration.

TABLE I
STATISTICAL ANALYSIS OF CONFIGURATION SETS PRODUCED BY DIFFERENT SAMPLING ALGORITHMS

Sampler	ICl	$\mathcal{G}(\mathcal{F})$	NN-Jaccard	NN-Jaccard (p10)	Mean configuration size
Chvatal	63.0	0.23	0.52	0.25	159.61
ICPL	62.2	0.23	0.52	0.25	162.19
Yasa	15.8	0.07	0.70	0.64	268.74
IncLing	18.3	0.07	0.53	0.48	316.16

From Table I, Chvatal and ICPL exhibit the same feature-level Gini coefficient, and Yasa and IncLing share the same value as well. Moreover, Chvatal and ICPL produce more configurations than Yasa and IncLing; accordingly, the usage of minority features is comparable within the former pair. For configuration similarity, both NN-Jaccard metrics for Chvatal and ICPL are identical and lower than those for Yasa and IncLing. Notably, Yasa yields the highest NN-Jaccard and NN-Jaccard (p10), which aligns with the APFD pattern in Fig. 1b. These observations suggest a plausible mechanism. Higher similarity leads to greater overlap among Yasa’s configurations, so early selections introduce fewer new interactions or features. This conflicts with HCP’s novelty objective and, consequently, reduces fault-detection efficiency. Finally, IncLing attains the largest mean configuration size; as a result, each configuration tends to manifest more features and interactions, which likely contributes to its superior Rank.

a) *Discussion*: HCP attains its strongest fault-detection performance on configurations sampled by IncLing, followed by Chvatal and ICPL, and it performs worst with Yasa. A likely explanation, supported by our distributional analysis, is that Yasa produces configurations with higher mutual similarity, which weakens HCP’s ability to differentiate priorities via feature novelty and configuration diversity. By contrast, IncLing tends to yield larger configurations on average, which helps surface fault-triggering combinations earlier. These findings provide practical guidance for the design of sampling algorithms.

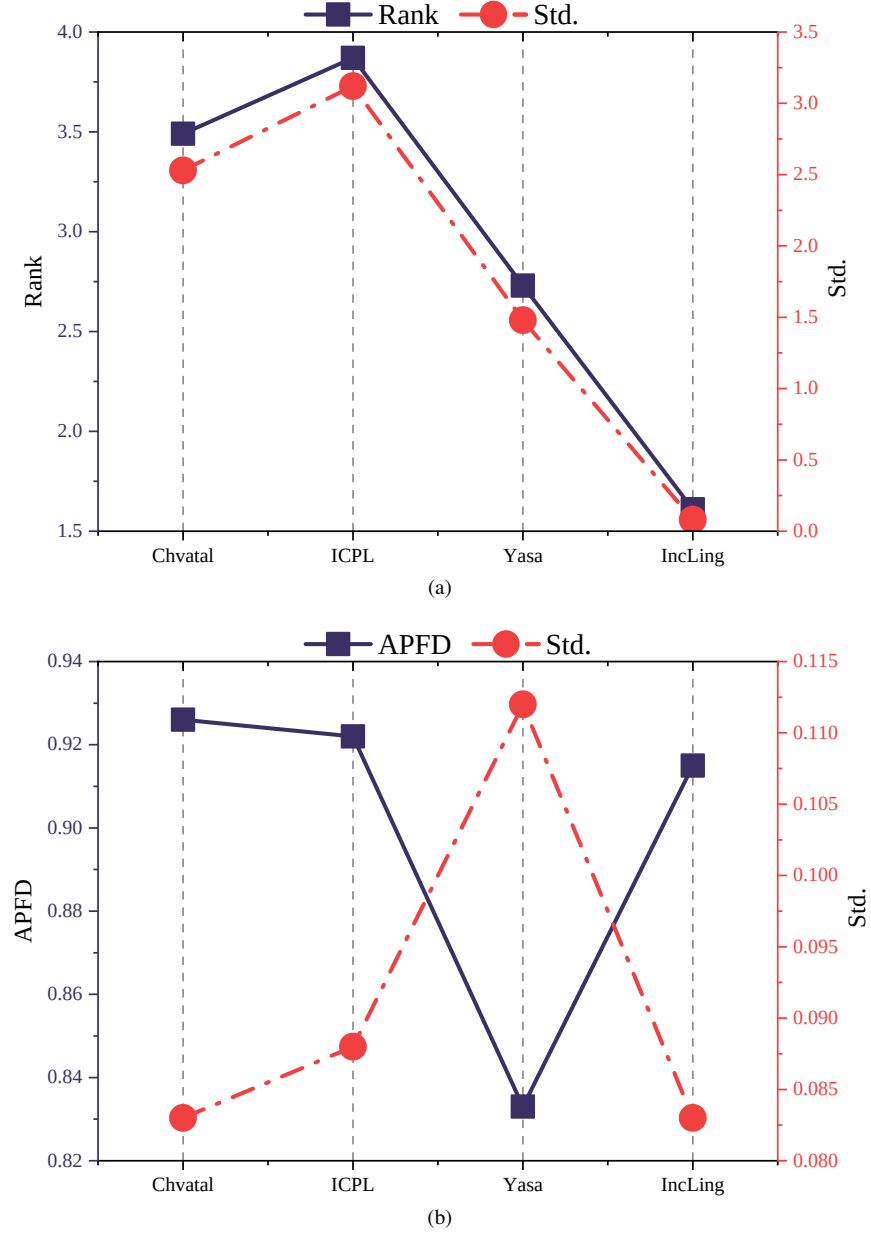


Fig. 1. Effect of sampling algorithms on fault-detection performance of HCP

Answer to RQ4

Therefore, the answer to this RQ is clear: greater diversity among configuration sets and larger average configuration sizes enhance HCP's fault-detection performance.

C. Ablation Study (RQ5)

Fig. 2 illustrates how different settings of a and b affect the APFD achieved by HCP. Notably, the weight assigned to diversity is determined by the choices of a and b . As shown in Figure 10, setting any one of the three weights to zero prevents HCP from attaining the best APFD; the diagonal entries in the plot correspond to a zero diversity weight. This observation underscores the necessity of jointly considering internal feature interactions, novelty, and diversity. In terms of values, interaction weights in the range 0.3~0.6, novelty weights in 0.2~0.4, and diversity weights in 0.1~0.2 tend to benefit HCP's fault-detection performance. However, setting the novelty weight to 0 has a larger adverse impact on APFD than setting the interaction weight to 0, and it markedly reduces APFD. The primary reason is that a greedy emphasis on a single element tends to cluster similar configurations early in the schedule, thereby diminishing fault-detection efficiency.

To further investigate how different components of the edge weight affect the fault-detection performance of HCP, Fig. 3 shows the APFD of four variants across different systems in FCM. HCP-w/o-IDF, HCP-w/o-DC, HCP-w/o-BE, and HCP-

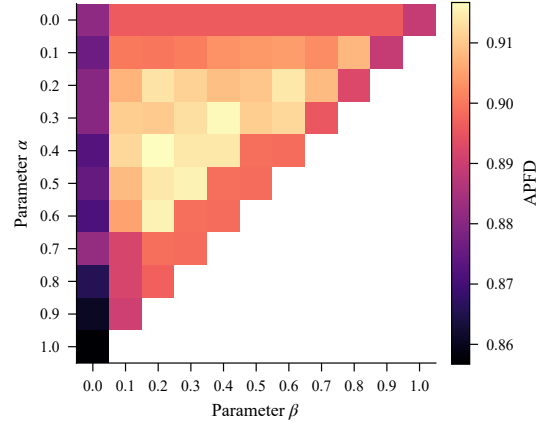


Fig. 2. Effect of varying α and β on the APFD of HCP.

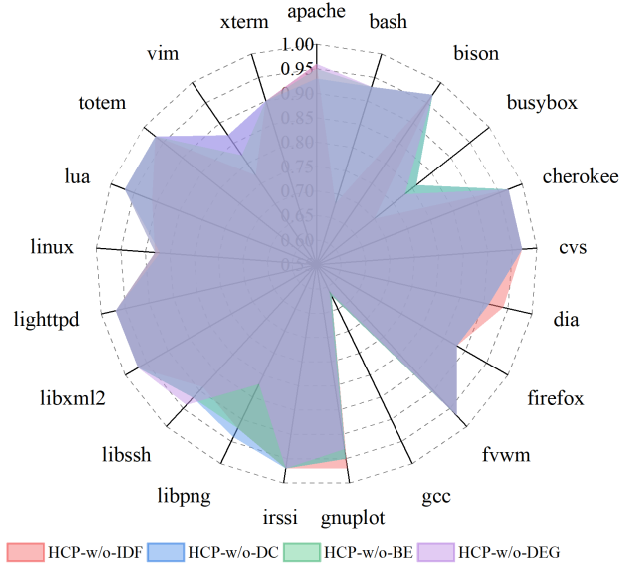


Fig. 3. APFD comparison of edge-weight components for HCP across 21 systems.

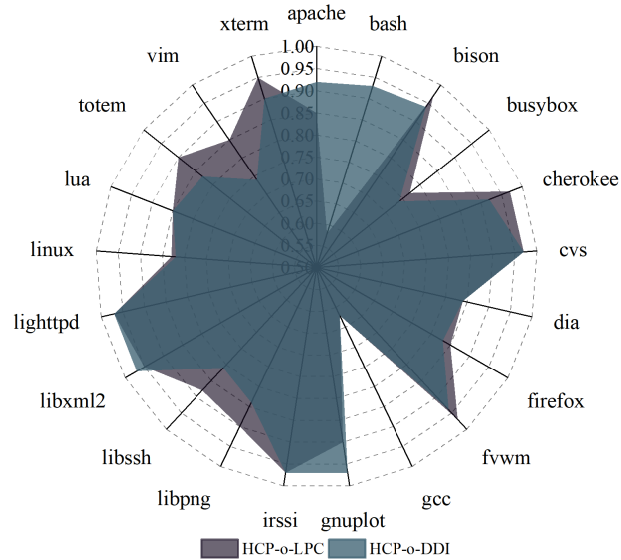


Fig. 4. APFD of HCP variants differing in the global edge set construction (control-only vs. data-dependence-only) across 21 systems.

w/o-DEG denote the variants that omit $IDF(\cdot)$, $DC(\cdot)$, $BE(\cdot)$, and $DEG(\cdot)$, respectively, when computing edge weights. From Fig. 3, disregarding edge rarity (HCP-w/o-IDF) has the largest overall impact on HCP in terms of APFD, particularly for `bash`, `busybox`, and `lua`. For the other components, their effects on HCP are not markedly different. On most systems, e.g., `bash`, `bison`, `cherokee`, `cvs`, `lua`, the other components yield similar APFD. These results indicate that each component influences the interaction-coverage score to some extent, thereby affecting HCP’s fault-detection performance. Because the 4 variants show performance advantages on different systems, this suggests that different parts of the edge-weight computation may better fit certain fault distributions. Therefore, it is meaningful to consider all 4 components jointly when computing edge weights.

Finally, Fig. 4 presents the fault-detection performance of two HCP variants: one constructs the global edge set using only control interactions (HCP-o-LPC), and the other uses only data-dependence interactions (HCP-o-DDI). Pronounced performance differences between HCP-o-LPC and HCP-o-DDI appear on several systems, including `apache`, `bash`, `libpng`, `libssh`, `totem`, `vim`, and `xterm`. On these systems, the two variants exhibit complementary strengths. For `apache` and `bash`, omitting control interactions substantially degrades performance; by contrast, on `totem`, `vim`, and `xterm`, data-dependence interactions have greater impact. This behavior accords with the fault distribution in FCM. Some systems are dominated by control-interaction faults, whereas others are dominated by data-dependence faults. Therefore, analyzing only a subset of interactions—as in COPRO—is insufficient. Overall, the results indicate that both control and data-dependence interactions should be considered simultaneously.

Answer to RQ[

title=Answer to RQ5] For RQ5, feature-interaction coverage and feature novelty are key contributing factors, while incorporating configuration diversity remains essential for maintaining balanced performance. Meanwhile, the influence of individual edge-weight components varies across systems; considering them jointly further improves the performance of HCP. Similarly, when constructing the global edge set, it is necessary to consider both control interactions and data-dependence interactions, allowing HCP to adapt to heterogeneous fault distributions.